# Anshika Mittal

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Portfolio | LeetCode | Codeforces

## Education

**Vellore Institute of Technology**
*B.Tech. Computer Science and Engineering (Specialization in E-Commerce Technology)* Sep 2023 – Ongoing
• Cumulative GPA: **9.46 / 10.0**

## Technical Skills

- **Programming Languages:** C++, Python, JavaScript
- **Tools & Frameworks:** Node.js, Express.js, React.js, Tailwind CSS, Git, Postman, AWS EC2, AWS S3, Mongo DB
- **Machine Learning:** Numpy, Pandas, Scikit-learn, Matplotlib, EDA

## Experience

**amasQIS.ai** Remote
*AI Intern* July 2025 – Dec 2025

- Implemented and fine-tuned supervised (Linear Regression, Logistic Regression) and unsupervised (K-Means) ML models in Python, improving classification accuracy by 12% across high-dimensional datasets.
- Engineered robust ETL (Extract, Transform & Load) and data pre-processing pipelines using Pandas, NumPy and Scikit-learn, reducing training data time by 20%.
- Conducted model evaluations using metrics such as accuracy, F1-score, precision and recall, identifying and resolving performance bottlenecks.

## Projects

**Scalable Notification Service [Github]** Jun 2026 - Aug 2026
*Technologies: Node.js, Express.js, MongoDB*

- Engineered a multi-channel notification processing system supporting Email, processing numerous queued notifications in seconds through asynchronous workers.
- Reduced notification API latency by decoupling delivery from request handling using Redis-backed asynchronous job queues.
- Prevented duplicate notification processing across 10+ repeated requests using database-level idempotency constraints and implemented exponential-backoff retries with DLQ handling for persistent failures.
- Reduced notification-history query latency by 85% through targeted MongoDB indexing while using Strategy and Factory patterns to decouple channel-specific delivery logic.

**Resilient Customer Churn Prediction Platform [Live Demo] [Github]** Jan 2026 – Jun 2026
*Technologies: Python, Scikit-learn, SHAP, React, Node.js, Express.js, MongoDB, AWS S3*

- Built a decoupled, explainable churn prediction platform to identify at-risk customers, achieving 80.81% accuracy across 30+ demographic, service and billing features.
- Improved model transparency by integrating SHAP LinearExplainer, translating individual prediction weights into interpretable feature-contribution visualizations for end users.
- Scaled batch inference to process 10,000-record CSV uploads while optimizing MongoDB operations to achieve a 15% reduction in write latency.
- Secured cloud-based profile image management with AWS S3 by enforcing file-type and 5MB size validation and implementing automated garbage collection to eliminate orphaned objects.

## Achievements

- NPTEL SWAYAM Deep Learning (IIT Ropar, Top 5%)
- SC-900: Microsoft Security, Compliance, and Identity Fundamentals
- NPTEL SWAYAM Marketing Management - II (Top 1%)
- Coursera Industrial IoT Markets & Security
- Ranked 4,495 out of 40,113 participants in LeetCode Weekly Contest 510
- Secured 100/100 in Mathematics for 3 consecutive years in classes X, XI & XII CBSE Board.
- Class XII (CBSE Board) - 95.8% Mar 2022 – Mar 2023
- Class X (CBSE Board) - 98.2% Mar 2020 – Mar 2021

## Extra Curriculars

Music (Flute) | Sport (Judo) | Language (Spanish) | Games (Chess)